# Tim Richardson

Software Engineer, Applied AI • Genomics and Health Data

London, UK | 07828134100 | tim@rewire.it | rewire.it | github.com/timini

## Summary

Software engineer with 15 years' experience, over a decade of it on genomic and health data. At Genomics England I work on national-scale genomic pipelines, build the Claude Code tooling for the organisation's AI coding pilot, and manage a team of six. At IQVIA I deployed patient-level prediction models on real-world health data. I build open-source tools for AI agents and test them with pre-registered experiments.

## Technical Skills

| | |
|---|---|
| **AI and ML** | LLM applications, RAG, agents and tool use, model deployment, evaluation and benchmarking; Anthropic, OpenAI and Gemini APIs, Google ADK |
| **AI engineering** | Claude Code skills and plugins, MCP servers, agent memory, mutation testing |
| **Genomics and health** | Nextflow, nf-core, VEP, Samtools, BWA, Singularity, LSF, FHIR, ontologies, real-world data |
| **Languages** | Python, Go, TypeScript, SQL |
| **Platform and data** | AWS, GCP, Docker, Kubernetes, Terraform, FastAPI, PostgreSQL, Elasticsearch, vector and graph databases, React |

## Applied AI Projects

**Agentic Verification** github.com/timini/agentic-verification

Claude Code plugin and GitHub Action that checks whether tests would catch broken code, by running every mutant of the lines a change touches. Evaluated in a pre-registered study and agent benchmarks on 8 open-source Python libraries, written up as a paper. On 14 real commits, the agent's tests caught 99% of mutants with the plugin against 92% without.

**neuromem** github.com/timini/neuromem

Long-term memory for LLM agents modelled on hippocampal consolidation, with providers for Gemini, OpenAI, Claude and local models. Scores 0.870 on the LongMemEval-s benchmark (n=100).

**drill-me** github.com/timini/drill-me

Claude Code plugin that turns Claude into an adaptive tutor, with spaced repetition and a memory of what you got wrong. 15 GitHub stars.

## Experience

**Genomics England** Dec 2021 – Present

*Senior Software Engineer and Performance Development Lead (line manager)* London, UK

- **Genomic pipelines:** core engineer on the migration of national-scale genomic workflow orchestration (100,000+ genomes) to Nextflow, for scalability and reproducibility.
- **Applied AI:** built Claude Code skills, plugins and MCP server integrations for the organisation's AI coding assistant pilot, and trained the pilot teams to use them.
- **Leadership:** line manager for 6 engineers; hired new team members and supported promotions.
- **Backend:** Python services on AWS for high-throughput genomic data processing.

**IQVIA** Mar 2019 – Dec 2021

*Senior Software Engineer, AI/ML Team* London, UK

Go, FastAPI, TypeScript, React, Elasticsearch, PostgreSQL, D3.js

- Deployed patient-level prediction models (cohort and outcome prediction on real-world health data) as FastAPI services behind the analytics product.
- Built the product's Go backend, React frontend and statistical features for real-world healthcare data.
- Coordinated releases, product development, process improvement and code review.

**Project Shivom** Oct 2018 – Mar 2019

*Lead Developer* London, UK

Google Genomics, Cloud Build, PostgreSQL, GraphQL, React Apollo, smart contracts

- Designed, built and deployed a genomic data sharing platform with blockchain components.

- Managed the development team; built the REST backend and React frontend; deployed with Docker and App Engine.

**Congenica** Mar 2016 – Jul 2018
*Senior Frontend Developer* Hinxton, UK
Perl, React, Redux, reactive programming, BioPerl

- Developed and optimised features of a clinical genomics tool with Perl APIs and a React frontend, focusing on data flow in a data-intensive UI.
- Led development of the API integrating Congenica's Sapientia platform with Genomics England's 100,000 Genomes Project.

**Repositive** Nov 2014 – Mar 2016
*Lead Developer* London, UK
Docker, Node.js, Ansible, Ember.js, PostgreSQL, Elasticsearch

- Led development of a genomic data discovery platform, including its infrastructure and development process.
- Designed the Node.js and Express API on Redis, PostgreSQL and Elasticsearch, with an Ember.js single-page frontend.

**Earlier roles**

| | | |
|---|---|---|
| **Polecat**, Python Contractor | Python reporting service generating PowerPoint reports (Django, GraphQL, React, Elasticsearch) | 2018 |
| **Wolfson Institute**, Freelance | ChIP-seq analysis: QC, peak calling, motif discovery, GO annotation | 2014 |
| **Fire Tech Camp**, Instructor | Taught Python to children aged 11 to 15 | 2014 |
| **Ixxy**, Django Developer | Sites on a Django CMS | 2013 – 2014 |
| **Ecomarket.com**, Django Developer | Backend for an online marketplace (Django, PostgreSQL) | 2013 |
| **Swerve Concepts**, Junior Developer | Full-stack Python and PHP; built Swerve Media Hub and SignUpAnywhere | 2011 – 2013 |

## Education

**Birkbeck, University of London** 2012 – 2014
*MSc Bioinformatics and Systems Biology (Merit)* London, UK
Statistics, protein structure, sequence alignment, systems biology and software development. Thesis: tracing human migration with MCMC population clustering of dense genotype data.

**University of Leeds** 2005 – 2008
*BSc Cognitive Science (Hons)* Leeds, UK
Computing, psychology and philosophy. Dissertation: data mining techniques for dynamic data sets.